Supplementary Information: Experiment-Driven Bayesian Optimization of the Impact Response of Multi-Material 3D-Printed Tensegrity-Inspired Structures

Draft note (remove before submission): as in the main text, any value printed in this violet color is a synthetic placeholder rather than a measurement. As of 2026-09-15 this Supplementary Information contains no violet values. The former placeholder batch-2 print key was replaced by the logged campaign record, which now covers batches 2 to 4.

This Supplementary Information collects the detailed design-, fabrication-, test-, and analysis-development records supporting the main manuscript. Unlike the main text, the SI references the project's development history directly, including specific pull requests (PRs) and issues in the project repository (https://github.com/vertical-cloud-lab/tensegrity-optimization).

S1 Joint-Design Study

We developed and ranked the strut-to-cable junction through a five-design OpenSCAD study (anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar variants). The working prototype uses a dovetail joint (Design B) with an anchor-bulb backup (Design A) for sensitivity studies. A captive TPU core is routed inside the PLA shell so that the soft tendon is protected from layer-line failure at the transition.

Full results, renders, and the Phase-3/Phase-4 design reviews are documented in PR #39 (joint design) and PR #35 (CAD parameterization and print history).

S2 Support Strategy for Soft Members

Near-vertical TPU tendons are otherwise unsupported during printing. For the printed campaign batches, we disabled automatic support generation and painted PLA edge supports manually in Bambu Studio along each tendon. We tightened the support-to-object XY clearance from 0.35 mm to 0.2 mm during tuning on the S0 reference prism (issue #98). Support removal is the leading source of the cosmetic defects that we log per article in the print key (Section S3). They are surface stringing and, on two seed articles, a TPU strand that did not fuse to a bottom tendon and came away from it.

In parallel, we developed an automated alternative. It combines a tensegrity-specific Bambu Studio recipe (support threshold angle lowered from 40° to 10°, support material matched to the rigid extruder) with manually generated narrowing pillars that ray-cast against the printable mesh underside. The narrowing-pillar workflow is documented in PR #66, and the H2D multi-part material assignment fix in PR #64.

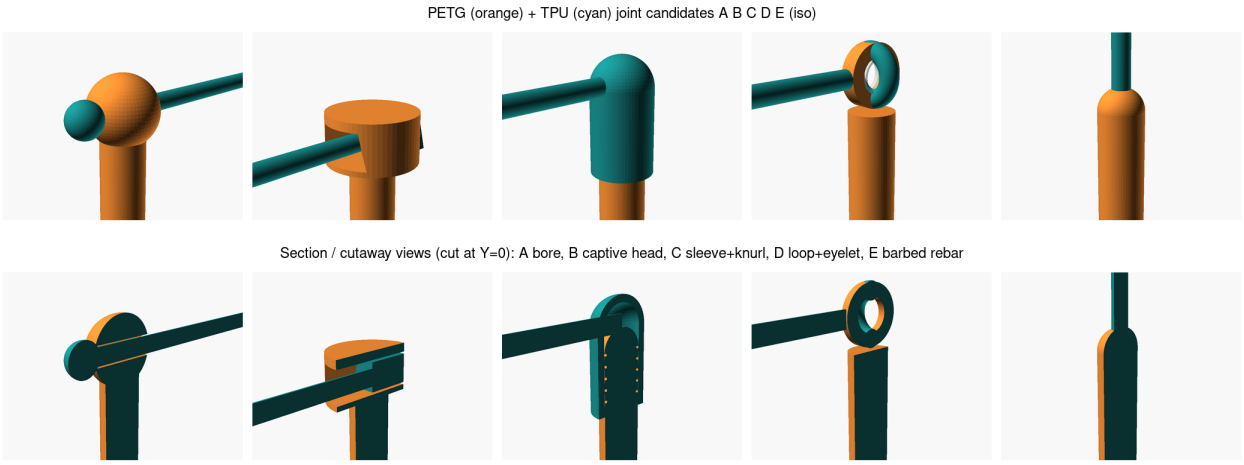


Figure S1: Five candidate strut-to-cable joint geometries from the OpenSCAD joint study (left to right: A anchor-bulb, B dovetail, C TPU-sleeve overmold, D eyelet-loop, E TPU-rebar; top row isometric, bottom row section views). The study renders use the polyethylene terephthalate glycol (PETG, orange) and TPU (cyan) material pair under which the joint family was first developed. The working prototype adapts the selected dovetail (B) with an anchor-bulb (A) backup to the PLA/TPU pair of the main text, with the captive TPU core routed inside the PLA shell.

S3 As-Printed Campaign Record (Print Key)

Table S1 is the print key for the seed campaign: every article's print identifier, its design, its weighed mass, and its logged defects. The authoritative machine-readable key (`bo/t3-prism-bo-batch-print-key.csv`) and the as-printed slicer project (one specimen per named build plate) live at PR #102. Two entries deserve care. We printed design 08 three times. The plate label in the archived slicer project and a contemporaneous issue comment disagree on which article is the official one (*dea4ls* versus *bag26v*), and pending confirmation we use *bag26v* in the analysis. The triplicate prints of design 08 also supply the only print-to-print repeatability number for a seed design, a mass standard deviation of 0.457 g. Three prints of a single geometry do not give a general process repeatability estimate. Separately, one tested article (*amdjwm*) has no confirmed design mapping and is excluded from the surrogate fit until identified.

Table S2 is the corresponding logged key for the three model-recommended batches. Every batch-2 and batch-3 article printed at about 11% dry-box relative humidity and every batch-4 article at about 7%. The batch-3 plate was printed three times. The second-print articles (2dran1 to 2dran9) weighed 19.48 to 20.32 g, on average 0.27 g heavier than the matching first-print articles, and were all logged defect-free. The third-print articles (dran31 to dran39, printed 2026-09-28) weighed 19.68 to 20.56 g (mean 20.07 g), and the print log notes no defects. The lab record confirms each 2dran$N$ and dran3N as a re-print of drran$N$ at the same Ax trial, so the three prints pair article for article (tabulated in the main-text Results). The nine second-print articles entered the surrogate fit as distinct completed trials at their own weighed masses and per-session standard errors. The third print was drop-tested on 2026-09-28, 2026-09-29, and 2026-10-02 (20 drops each, all captures clean) and has not yet entered the fit. Its records are on PR #86. The authoritative machine-readable keys, with plate positions and the mapping evidence per row, are the `t3-prism-bo-round1/round3/round4-print-key.csv` files archived with the manuscript data.

Table S1: Print key for the seed campaign (nine Sobol designs plus the S0 reference). Masses weighed on a 0.01 g scale. TPU fed from a heated dry box near 8% relative humidity where logged. The first five articles (printed before the TPU nozzle fix) used the standard 0.4 mm TPU nozzle, and the rest used the high-flow 0.6 mm nozzle.

Design	Article	Mass (g)	Logged defects
Spec 00	9hhbkp	21.62	none
Spec 01	6lhxfy	18.50	slight strut misprint; divot on one top tendon
Spec 02	autv5r	22.04	minor tendon misalignment near top
Spec 03	ebdna8	19.61	two strands separated from bottom tendons (support removal)
Spec 04	nvxsrv	20.66	misalignment ring after printer pause
Spec 05	6nheas	21.73	two strands separated from bottom tendons (support-related)
Spec 06	1zm8rv	18.50	stringing along diagonal tendons
Spec 07	ajhby6	20.53	filament line pulled from bottom struts
Spec 08	dea4ls	22.29	stringing; residual PLA on tendon
Spec 08	bag26v	21.42	some stringing on tendon
Spec 08	ghmj4y	22.10	inconsistent tendon cross-section near top
S0 ref.	bpx68c	20.23	none major

S4 Drop-Tower Protocol and Rig Characterization

The campaign fixture is a Lansmont Model 23 shock test system with a Test Partner 4 data-acquisition console. This section summarizes the development history behind the protocol in the main text.

Protocol. The drop height is 60 in, with the specimen upright on the carriage plate. We discard the first two drops of each session as seating warm-up. Session length differs by batch. Seed sessions contribute 101 valid captures per article (103 for autv5r), the batch-2 record lists 21 to 22 valid captures per article, and batch-3 and batch-4 sessions recorded 20 drops per article. Exact per-session counts live in the archived per-drop records, whose two generations differ on whether the count is quoted before or after the warm-up discard. Section 3.3 of the main text justifies the shortening. The statistical minimum from the pre-campaign variance study is five stabilized drops, and a truncation replay of the seed sessions preserves the full-session transmissibility ranking at twenty drops. The three prints of batch 3 also showed that the error budget is dominated by seating and print variation rather than session length. We record three high-speed video clips (959 fps) per specimen. From batch 4 on, we film them at drops 1, 10, and 20 so that clips pair deterministically with captures. Testing takes roughly 42 s per drop, so about ninety minutes per seed article and fifteen per later-batch article. Capture settings: 1.25 MHz sample rate, 100 ms record, 2 ms pre-trigger, 150 G trigger on the base channel, AC-coupled ICP conditioning. After a settings-loss event, we re-verified the console settings digit-for-digit against an archived screenshot and confirmed data continuity to within about 1%. Per the review instruction of 2026-09-02, the per-session capture counts above should be re-verified by a person against the committed per-drop records before submission.

Impact surface. A blinded 3×3 crossover experiment over three candidate polyurethane mat arrangements (single 0.25 in Shore 70 OO, single 0.5 in Shore 60 OO, stacked) selected the single 0.5 in Shore 60 OO sheet (issue #88, PR #86). Unlike the earlier felt stack, the mat does not measurably compact over hundred-drop sessions. Sessions recorded as “5 felts” in early protocol

Table S2: Print key for the model-recommended batches, first prints only. The text summarizes the second and third prints of the batch-3 designs. Masses weighed on a 0.01 g scale.

Batch	Trial	Article	Mass (g)	Logged defects
2	10	r2d2c9	17.91	none
2	11	r2d2c4	19.33	string on one top tendon; partial TPU detachment
2	12	r2d2c2	17.96	none
2	13	r2d2c1	19.38	detached TPU string, two top tendons; odd foot support
2	14	r2d2c6	19.11	none
2	15	r2d2c7	18.51	none
2	16	r2d2c3	18.63	none
2	17	r2d2c8	19.98	full-span strings, two bottom tendons
2	18	r2d2c5	23.47	none
3	28	drran5	19.88	tiny bubbles, diagonal tendons
3	29	drran4	19.92	none
3	30	drran9	19.34	none
3	31	drran3	19.92	tiny bubbles, diagonal tendons
3	32	drran7	20.06	tiny bubbles, diagonal tendons
3	33	drran2	19.33	none
3	34	drran6	19.24	tiny bubbles, diagonal tendons
3	35	drran8	19.32	none
3	36	drran1	19.31	none
4	37	corny7	19.62	none
4	38	corny2	20.21	none
4	39	corny8	19.64	none
4	40	corny5	20.58	none
4	41	corny6	19.45	none
4	42	corny1	20.20	TPU mixed into one PLA strut layer
4	43	corny9	19.46	none
4	44	corny4	20.56	small pores, diagonal tendons
4	45	corny3	20.15	none

documents were physically four felt sheets plus one cardboard layer.

Sensors and calibration. Base input: single-axis piezoelectric accelerometer on the carriage plate (1.059 mV/G, full scale 9443 G). Top response: Dytran 3133A4 triaxial accelerometer (X/Y/Z 0.690/0.667/0.734 mV/G) seated in a printed pocket at the top vertex, mounted per ISO 5348. We retired an earlier single-axis channel after range clipping. We cross-calibrated the two base-capable channels in place over fifteen co-located drops. The ratio was 0.953 ± 0.001 at CFC-180 (per-drop 0.952 ± 0.004) and stable across drop heights (PR #74).

Filtering correction. We implemented the SAE J211/1 channel-frequency-class filter used for all reported numbers from the standard's published coefficients. An earlier in-house implementation passed the two-pass (forward-backward) corner frequency to a single-pass design, which narrows every channel class by roughly 20% (an effective CFC-144 where CFC-180 was intended). We identified the discrepancy during the analysis audit of issue #94 and corrected it before the campaign

numbers were produced.

Processing pipeline. For each capture, we calibrate and CFC-filter every channel independently. We form the top-vertex magnitude $\|\mathbf{a}_{\text{top}}(t)\|$ from the three filtered triaxial components and search for the two peaks entering t_{180} within a short window (approximately 1.5 ms) of first contact, so the peaks need not be simultaneous. We estimate the baseline from the record tail, because at healthy tower speed the mat-contact foot begins before the 2 ms pre-trigger window. Δv is the integral of the baselined base channel over the impact pulse and doubles as the rig-health check. Drops are discarded per the warm-up rule. Table S3 collects the operational analysis parameters in one place. Values known from the protocol are stated numerically. The remaining thresholds are to be transcribed verbatim from the version-tagged campaign analysis code at its archival release, so that the table matches the code of record.

Table S3: Operational analysis parameters for the drop-tower pipeline. "Code of record" marks values to be transcribed from the version-tagged analysis release.

Parameter	Value
Drop height	60 in (1.524 m)
Sample rate / record / pre-trigger	1.25 MHz / 100 ms / 2 ms
Trigger	150 G on the base channel
Reporting filter class	CFC-180 (SAE J211/1), two-pass corner corrected
Diagnostic filter class	CFC-1000
Peak-search window	≈1.5 ms from first contact
Baseline estimate	record tail
Warm-up exclusion	first 2 drops per session
Captures per session	101 seed (103 autv5r); 21-22 batch 2; 20 batches 3-4
Rig-health check	Δv from baselined base channel
Ringdown fit gate	$r^2 \geq 0.85$ (exponential fit)
Second-contact threshold and window	code of record
Per-capture exclusion rules	code of record
Stabilized-mean convention	code of record
Filter initialization / boundary handling	code of record

Rig health and repair history. We check rig health against the impact velocity change Δv recovered from the base channel. Over an uninterrupted 100-drop session the Δv coefficient of variation (CV, the standard deviation divided by the mean) was 0.54% with a flat drift slope. Within a session, Δv and the CFC-180 input typically ease down by 1 to 4% as the polyurethane mat warms, and recover fully after a rest of minutes. After a mat rest or re-seat, the first 5 to 10 drops carry the settling transient, which is why the warm-up discard exists. The arrest velocity has visibly stabilized as cumulative drop count grew, so we keep the Δv record. During commissioning, two guide-rod locking pins failed, and we discovered and recommissioned a defeated pin-park safety interlock. The pin-related friction produced a measurable Δv deficit that resolved after repair (issue #92). We have performed more than 4,400 instrumented drops across roughly 80 printed specimens over about four months of commissioning, characterization, and campaign sessions (tallied through issue #101 plus the batch records). The campaign batches themselves contribute about 1,500 stabilized captures including the second print of batch 3, and its third print adds 162 more.

Seating sensitivity, the between-seat rule, and the seat gauge. The three prints of batch 3 (main text, Section 4.2) showed that a session can be internally stable and still misrepresent its design. The 1.251 session passed every within-session check, and the same design's two later prints read 1.029 and 1.045. Two standing conventions followed. First, under the *between-seat replication rule*, any single-seating value that would drive a decision requires confirmation on an independent re-seat. Second, we defined an advisory *seat gauge*, the broadband ratio of CFC-1000 to CFC-180 transmissibility. It reads 1.00 to 1.07 on most sessions, and we treat a session reading 1.15 or more as a re-seat candidate. The third print is the first replicate data collected after the gauge was defined (2026-09-12). Its one gross outlier (dran35, 10.1% above its design's median over the three prints) read 2.43, and its other eight sessions read 1.01 to 1.06. Over all 27 batch-3 sessions, the four reading 1.15 or more were each their design's highest value, and the gauge correlates with the deviation from the design median (Spearman $\rho = 0.52$, $p = 0.006$). Both conventions, with the session-by-session drift-watch record, live in the drop-test datasets on PR #86. The three-print comparison is archived with the manuscript data (`t3-prism-bo-round3-three-print.csv`, generated by `scripts/figures/round3_three_print_analysis.py`).

Objective provenance and adversarial review. The objective stack evolved through a measurability review (PR #97). Per-specimen crush energy in joules is not recoverable from this rig, because it has no falling-mass or displacement channel and the specimens stay elastic. The campaign responses are therefore the filtered peak-acceleration ratio and a provisional velocity loss index, optimized in a mass-weighted form, with quasi-static crush metrics deferred to a load-frame campaign. We pre-registered an adversarial review of the objective choices with an external AI referee (Edison ANALYSIS task `3e398131`, archived in the repository). It recommended accounting for article-level rather than per-drop noise, treating the velocity loss index as a candidate constraint pending a restrained-versus-unrestrained validation with synchronized video, and reserving print capacity for replicate articles. The main text records how the campaign responded to these recommendations. We printed and tested the batch-3 plate three times, which measured the print-to-print scatter directly and produced the between-seat replication rule. The velocity loss index remains a candidate constraint, and the surrogate continues to ingest per-session standard errors, with the replication evidence carried as an interpretation bound rather than as an inflated noise term. We plan to act on the reserved-replicate recommendation through the replicate rank-stability study of the main-text Methods. That study will re-print the best-, mid-, and worst-predicted in-sample designs in nominal triplicate and test whether the predicted ordering survives print-to-print and seating variation.

S5 Printed-Mass Model

We projected the seed batch onto constant *solid* CAD mass ($m^\star = 30.95\,\mathrm{g}$), yet the articles weigh 18.50 g to 22.29 g. The gap arises because PLA struts print at partial infill (15% grid, two walls) while thin TPU cables print near solid, and the PLA-to-TPU volume split swings widely across the design box. A two-stage calibrated model closes this gap (PR #102, `bo/t3_prism_mass_model.py`). Analytic member volumes are corrected to rendered-solid grams (one factor per material). Rendered-solid grams are then mapped to weighed grams through a wall-plus-infill term in the as-printed strut diameter (fitted infill 23.6%, wall 1.18 mm, TPU factor 0.744). The model's residual standard deviation is 0.378 g. For context, the three available prints of design 08 have a sample standard deviation of 0.457 g. Three prints of one geometry are insufficient to establish a general process repeatability limit. Run in reverse inside the

projection, the model holds *printed* grams constant without running the slicer. For a target mass, the projection finds the uniform scale factor at which the predicted printed mass equals the target, and we print the design at that scale. Bisection is enough for this, because predicted mass rises monotonically with scale. Batches 3 and 4 used a target of 20.23 g, the mass of the printed reference article. Batch mass CV fell from 5.5% (tested seed articles) and 8.7% (batch 2, still solid-mass-projected) to 1.7% (batch 3) and 2.3% (batch 4). The weighings sit at or below the model's estimate as a per-print flow offset (-0.52, -0.25, and -0.04 g for the three prints of batch 3 and -0.26 g for batch 4). In a first test of the model's infill terms, regressing the print-averaged mass residuals of the eighteen batch-3 articles on the two commanded infills gives slopes indistinguishable from zero over the realized 13 to 34% spread. We will recalibrate the session term with the accumulated weighings before the next batch's projection. Two batch-2 recommendation sets exist in the record. The batch actually sliced and sent to the printer was generated before this correction, under the seed batch's solid-mass projection. A subsequently regenerated set applies the constant-printed-mass projection and seeded the batch-3 tooling. Both are archived with the manuscript data (`round2-as-printed-solid-mass-projection.csv`, `round3-candidate-constant-printed-mass.csv`).

S6 Metal-Analog Validation Plan

A planned validation campaign will rebuild a subset of the T_3-prism designs as metal analogs, using the same R, H, θ, and member-diameter parameterization as the printed PLA/TPU specimens. The analogs will use hollow aluminum rods for the compression members and stainless-steel threaded cables for the tension members. To probe transfer across material systems rather than only confirming the optimum, the selected designs will deliberately span the predicted performance range. Representatives the surrogate predicts to be worst-, mid-, and best-performing will each be built in both material systems and characterized under the identical drop protocol of Section S4. The main-text metal-analog table and figure will report the measured PLA/TPU ranking and the hollow-aluminum/stainless-steel ranking on the primary campaign metric t_{180}, with the velocity loss index and, later, quasi-static specific energy absorption (SEA) as secondary rankings. Agreement between the two primary rankings would be preliminary evidence consistent with transfer beyond the as-printed polymers. Disagreement would not isolate which material or joint difference caused the change.

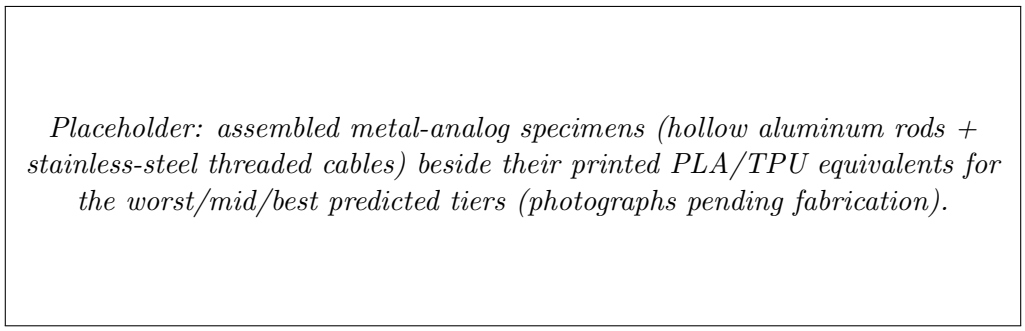


Figure S2: Metal-analog validation specimens alongside their printed PLA/TPU equivalents for each predicted performance tier.